

# Reference Sheet: Metrics, Algorithms & Definitions for Diversity in RAG Retrieval

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## 1 Notation

|                          |                                                      |
|--------------------------|------------------------------------------------------|
| $D$                      | Full dataset/candidate pool of $n$ items (chunks)    |
| $S \subseteq D$          | Selected subset, $ S  = k$                           |
| $\mathbf{e}_i$           | Embedding vector of item $i$                         |
| $u_i$                    | Utility (relevance) of item $i$ to the query         |
| $w_{ij} \geq 0$          | Similarity between items $i$ and $j$                 |
| $\cos(a, b)$             | Cosine similarity between vectors $a$ and $b$        |
| $\lambda, \alpha, \beta$ | Trade-off parameters between relevance and diversity |
| $T_{\max}$               | Context budget (max tokens in selected chunks)       |

## 2 Quick Reference

### Diversity Definitions

| Concept                      | Scope         | Approx. guarantee  | Key property                                    |
|------------------------------|---------------|--------------------|-------------------------------------------------|
| Max-min Dispersion           | Selected only | None (NP-hard)     | Maximises closest-pair distance in $S$          |
| Coverage (Facility-Location) | All of $D$    | $(1 - 1/e)$ greedy | Monotone submodular; respects data distribution |
| Combined Objective $F$       | All of $D$    | $(1 - 1/e)$ greedy | Modular utility + submodular coverage           |

### Selection Algorithms

| Algorithm                | Diversity type            | Relevance | Complexity    | Guarantee   |
|--------------------------|---------------------------|-----------|---------------|-------------|
| Top- $k$                 | None (baseline)           | Yes       | $O(n \log k)$ | —           |
| MMR                      | Dispersion (local window) | Yes       | $O(knw)$      | —           |
| gMMR                     | Dispersion (centroid)     | Yes       | $O(kn)$       | —           |
| FPS                      | Dispersion (global)       | No        | $O(kn)$       | —           |
| Greedy Facility-Location | Coverage                  | Yes       | $O(kn)$       | $(1 - 1/e)$ |
| Branch-and-Bound         | Coverage                  | Yes       | Exp. in $k$   | Optimal     |

### Retrieval Quality Metrics

| Metric                  | Range      | Requires            | What it measures                                                   |
|-------------------------|------------|---------------------|--------------------------------------------------------------------|
| Hit Rate (hit)          | $\{0, 1\}$ | Gold aliases        | Binary: does context contain the answer?                           |
| Facility-Location (fac) | $[0, 1]$   | Embeddings          | Average representativeness of $S$ over $D$                         |
| Recall@ $K$             | $[0, 1]$   | Relevance judgments | relevant retrieved / total relevant                                |
| Precision@ $K$          | $[0, 1]$   | Relevance judgments | relevant retrieved / (relevant retrieved + non-relevant retrieved) |
| nDCG@ $K$               | $[0, 1]$   | Graded judgments    | Ranking quality with position discount                             |
| $F_1$                   | $[0, 1]$   | Ground-truth answer | Token-level precision/recall of generated answer                   |
| DocRec                  | $[0, 1]$   | Gold documents      | Fraction of supporting documents selected                          |
| FactRec                 | $[0, 1]$   | Gold facts          | Fraction of supporting sentences selected                          |
| AvgCos                  | $[0, 1]$   | Embeddings          | Average query-chunk cosine similarity                              |

### Selection Overlap Metrics

| Metric             | Range    | What it measures                                                                                       |
|--------------------|----------|--------------------------------------------------------------------------------------------------------|
| Jaccard similarity | $[0, 1]$ | Set overlap between two selection strategies; $J \approx 1$ means the diversity method changes nothing |
| GoldConcat         | $\geq 0$ | Number of distinct answer aliases present in the concatenated context                                  |
| Gold/Chunk         | $\geq 0$ | Average answer-alias density per selected chunk                                                        |

### 3 Diversity Definitions

#### 3.1 Dispersion

Diversity measured by distances *between selected items only*. Does not consider unselected items.

**Max-min dispersion:**

$$\text{Div}(S) = \min_{i \neq j \in S} \text{dist}(i, j)$$

Maximizing this is NP-hard (the  $k$ -dispersion problem). No polynomial-time approximation guarantee for the greedy algorithm.

**Intuition.** Push selected points as far apart from each other as possible. The “weakest link” (closest pair) defines the score. Natural but ignores the structure of unselected data—may place representatives in empty regions of the space while leaving dense clusters uncovered.

#### 3.2 Coverage (Facility-Location)

Diversity measured by how well  $S$  represents the entire dataset  $D$ .

$$g(S) = \sum_{i \in D} m_i(S), \quad m_i(S) = \max_{j \in S} w_{ij}$$

- $m_i(S)$ : how well item  $i \in D$  is covered (similarity to its nearest representative in  $S$ ).
- $g(S)$ : total coverage over all items in  $D$ .

$g$  is **monotone submodular**:  $A \subseteq B \Rightarrow g(A) \leq g(B)$ , and for all  $A \subseteq B$ ,  $x \notin B$ :

$$g(A \cup \{x\}) - g(A) \geq g(B \cup \{x\}) - g(B) \quad (\text{diminishing returns})$$

**Intuition.** Every point in  $D$  should have a close representative in  $S$ . If 90% of the data clusters in region  $A$ , coverage allocates more representatives there. It respects the data distribution, unlike dispersion which may ignore dense clusters. Key advantage: submodularity gives provable approximation guarantees.

#### 3.3 Combined Objective

$$F(S) = \lambda \sum_{i \in S} u_i + (1 - \lambda) g(S)$$

Sum of a modular function (utility) and a submodular function (coverage).  $F$  is monotone submodular. Greedy gives a  $(1 - 1/e) \approx 0.632$  approximation.

**Intuition.** Trade off between picking the most relevant chunks ( $\lambda = 1$ , standard top- $k$ ) and picking chunks that collectively represent the whole candidate pool ( $\lambda = 0$ , pure coverage).

### 4 Selection Algorithms

#### 4.1 Top- $k$ by Relevance (Baseline)

Select  $k$  items with highest  $u_i = \cos(\mathbf{e}_q, \mathbf{e}_i)$ , subject to context budget  $T_{\max}$ .  $O(n \log k)$ .

**Intuition.** The simplest strategy: rank by similarity to the query, take the top. Fast but blind to redundancy—if the top chunks all say the same thing, the context is wasted.

#### 4.2 MMR (Maximal Marginal Relevance)

Greedy, dispersion-based. At each step, select:

$$c^* = \arg \max_{i \in R} \left[ \alpha \cdot \cos(\mathbf{e}_q, \mathbf{e}_i) - (1 - \alpha) \cdot \max_{j \in W} \cos(\mathbf{e}_i, \mathbf{e}_j) \right]$$

- $R = C \setminus S$ : remaining candidates.
- $W \subseteq S$ : context window (last  $w$  selected items).
- $\alpha \in [0, 1]$ : relevance–diversity trade-off.

First term rewards relevance; second penalizes similarity to already-selected items.  $O(k \cdot n \cdot w)$  per query.

**Intuition.** “Pick the next chunk that is relevant but not redundant with what I already have.” The penalty is against the single most similar already-selected item (pairwise max). The context window  $w$  limits computation and makes the penalty local—only the  $w$  most recent selections matter.

### 4.3 gMMR (Geometric MMR, from DF-RAG)

Greedy, dispersion-based. Replaces pairwise max with centroid distance, cosine with Euclidean.

$$\text{gMMR}(c) = \lambda \cdot \cos(\mathbf{e}_q, \mathbf{e}_c) + (1 - \lambda) \cdot \sqrt{2 - 2 \cos(\mathbf{e}_c, \mathbf{e}_{c_S})}$$

where the centroid of the selected set is:

$$\mathbf{e}_{c_S} = \frac{1}{|S|} \sum_{s \in S} \mathbf{e}_s$$

At each step:  $c^* = \arg \max_{c \in C \setminus S} \text{gMMR}(c)$ .

**Intuition.** Instead of penalizing similarity to the nearest selected item, gMMR rewards distance from the *center of mass* of all selected items. This is a more global view of diversity: “go far from where I’ve already concentrated,” not just “avoid my nearest neighbor.” Euclidean distance on normalized embeddings is  $\sqrt{2 - 2 \cos}$ , grounding diversity in geometric space.

### 4.4 FPS (Farthest Point Sampling)

Greedy, pure dispersion (no relevance component).

$$c^* = \arg \max_{i \in R} \left[ \max_{j \in S} \text{dist}(i, j) \right]$$

**Intuition.** At each step, pick the point that is farthest from all already-selected points. Pure space-filling strategy with no relevance signal. From 3D computer vision (point cloud subsampling). Slower than MMR because it uses Euclidean distance over all selected items rather than cosine over a window.

### 4.5 Greedy for Facility-Location Coverage

Start with  $S = \emptyset$ . Repeat  $k$  times: add  $j^* = \arg \max_{j \notin S} \Delta(j \mid S)$ , where the marginal gain is:

$$\Delta(j \mid S) = \lambda \cdot u_j + (1 - \lambda) \sum_{i \in D} \max(0, w_{ij} - m_i(S))$$

- Second term counts only the *new* coverage  $j$  brings.
- $m_i(S) = \max_{s \in S} w_{is}$ : current coverage of item  $i$ .
- $O(kn)$  per query (update  $m_i$  array incrementally).
- Provides a  $(1 - 1/e)$  approximation guarantee since  $F$  is monotone submodular.

**Intuition.** Each candidate is evaluated by: (1) its relevance  $u_j$ , and (2) the *new* coverage it adds—only counting items in  $D$  that it covers *better* than the current best representative. As  $S$  grows, the marginal coverage of each new item diminishes (submodularity), naturally preventing redundant selections.

## 4.6 Exact Branch-and-Bound for Coverage

Finds the optimal  $S^* = \arg \max_{|S|=k} F(S)$  exactly.

1. Nodes are partial sets  $S_p$ ,  $|S_p| \leq k$ . Priority queue ordered by upper bound.
2. **Upper bound** (exploits submodularity):

$$\text{UB}(S_p) = F(S_p) + \sum_{j \in R_{(1:t)}} \Delta(j \mid S_p)$$

where  $R_{(1:t)}$  are the  $t = k - |S_p|$  largest marginals among candidates.

3. **Lower bound**: any feasible completion (e.g., greedy from  $S_p$ ).
4. **Pruning**: if  $\text{UB}(S_p) \leq \text{BestValue}$ , discard node.
5. **Branching**: expand  $S_p$  by adding each candidate  $e \in D \setminus S_p$ , sorted by descending marginal gain.

Worst-case:  $O\left(\sum_{i=0}^k \binom{n}{i} \cdot n^2\right)$ . Polynomial in  $n$  for fixed  $k$ , but exponential in  $k$ . Practical for small  $k$  with effective pruning.

**Intuition.** The greedy solution is near-optimal but not exact. Branch-and-bound explores the combinatorial space of possible selections, using submodularity-derived upper bounds to prune most of the search tree. The key insight: submodularity guarantees that the sum of the  $t$  largest single-item marginal gains is an upper bound on the gain from adding *any*  $t$  items, because marginal gains can only decrease as the set grows.

## 5 Retrieval Quality Metrics

### 5.1 Hit Rate ( $\text{hit}_b$ , $\text{hit}_c$ )

$$\text{hit}(q) = \begin{cases} 1 & \text{if any gold alias of } q \text{ appears in } \text{concat}(S_q) \\ 0 & \text{otherwise} \end{cases}$$

Averaged over queries:  $\text{hit}_b$  (baseline),  $\text{hit}_c$  (coverage).  $\Delta\text{hit} = \text{hit}_c - \text{hit}_b$ .

**Intuition.** Binary pre-LLM recall: did the selected chunks contain the answer at all? If  $\text{hit} = 0$ , no LLM can produce the correct answer from this context.

### 5.2 Facility-Location Coverage Score ( $\text{fac}$ )

$$\text{fac}(S) = \frac{1}{|D|} \sum_{i \in D} \max_{j \in S} \cos(\mathbf{e}_i, \mathbf{e}_j)$$

$\Delta\text{fac} = \text{fac}_c - \text{fac}_b$ .

**Intuition.** Average per-item representation quality. High  $\text{fac}$  means every candidate chunk has a close representative among the selected chunks. Measures whether coverage-aware selection actually achieves better representativeness.

### 5.3 Recall@K

$$\text{Recall@K}(q) = \frac{|\{p \in \text{top-}K : p \text{ is relevant}\}|}{|\{p \in \text{corpus} : p \text{ is relevant}\}|}$$

**Intuition.** Fraction of all relevant passages captured in the top- $K$ . Used with TREC qrels where human relevance judgments are available.

## 5.4 nDCG@K (Normalized Discounted Cumulative Gain)

$$\text{DCG@K} = \sum_{i=1}^K \frac{2^{\text{rel}_i} - 1}{\log_2(i + 1)}, \quad \text{nDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}$$

$\text{rel}_i$ : graded relevance at rank  $i$ . IDCG: DCG of the ideal ranking.

**Intuition.** Not just *whether* relevant passages are retrieved, but *how well they are ranked*. Highly relevant passages at early ranks score much more than at later ranks (logarithmic discount). Normalized to  $[0, 1]$  by the ideal.

## 5.5 $F_1$ Score (DF-RAG)

$$P = \frac{|\text{tokens}(\hat{a}) \cap \text{tokens}(a^*)|}{|\text{tokens}(\hat{a})|}, \quad R = \frac{|\text{tokens}(\hat{a}) \cap \text{tokens}(a^*)|}{|\text{tokens}(a^*)|}, \quad F_1 = \frac{2 \cdot P \cdot R}{P + R}$$

where  $\hat{a}$  is the generated answer and  $a^*$  is the ground truth; tokens are compared as bags of words.

**Intuition.** Balances how much of the gold answer the system produced (recall) with how much of the system’s output is correct (precision). Used across all DF-RAG benchmarks for end-to-end QA evaluation.

## 5.6 DocRec / FactRec (HotpotQA)

$$\text{DocRec}(q) = \frac{|S_q \cap \text{gold\_docs}(q)|}{|\text{gold\_docs}(q)|}$$

**Intuition.** Did we select chunks from the *right* documents? In HotpotQA (2 gold docs out of 10), DocRec = 1.0 means both supporting documents are represented.

$$\text{FactRec}(q) = \frac{|\text{selected\_facts}(q) \cap \text{gold\_facts}(q)|}{|\text{gold\_facts}(q)|}$$

**Intuition.** Finer than DocRec: even if the right document is chosen, did we capture the specific *sentences* annotated as supporting facts? This is the most direct measure of whether diversity helps recover complementary evidence.

## 5.7 AvgCos (Average Cosine Similarity)

$$\text{AvgCos}(q) = \frac{1}{|S_q|} \sum_{j \in S_q} \cos(\mathbf{e}_q, \mathbf{e}_j)$$

**Intuition.** Average relevance of selected chunks to the query. Lower AvgCos means the strategy sacrificed relevance to gain diversity. Useful for diagnosing whether a diversity method is “paying too much” in relevance.

# 6 Selection Overlap Metrics

## 6.1 Jaccard Similarity

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Applied at chunk level ( $\text{Jaccard}_{\text{chunks}}$ ) or document/domain level ( $\text{Jaccard}_{\text{domains}}$ ).

**Intuition.** Measures agreement between two selection strategies.  $J \approx 1$ : the diversity method barely changes the selection (no effect).  $J \ll 1$ : genuinely different selections. The most diagnostic metric for determining whether a diversity method is actually doing something.

## 6.2 Gold-Alias Metrics

$$\text{GoldConcat}(q) = |\{a \in \text{aliases}(q) : a \in \text{concat}(S_q)\}|$$

**Intuition.** How many distinct forms of the answer appear in the selected context. Richer than binary hit rate—captures answer thoroughness.

$$\text{Gold/Chunk}(q) = \frac{1}{|S_q|} \sum_{j \in S_q} |\{a \in \text{aliases}(q) : a \in \text{chunk}_j\}|$$

**Intuition.** Average answer density per chunk. High values mean each chunk independently contains answer evidence; low values mean evidence is spread thinly.

## 7 Key Experimental Parameters

| Parameter          | Description                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------|
| $T_{\max}$         | Context budget in tokens. Tighter budgets force harder selection and amplify diversity effects. |
| $\lambda / \alpha$ | Relevance–diversity trade-off.                                                                  |
| $w$                | Context window size (MMR): number of recent selections considered for the diversity penalty.    |
| chunk_tokens       | Chunk size in tokens for document splitting.                                                    |
| stride             | Overlap between consecutive chunks (stride < chunk_tokens implies overlap).                     |
| max_docs           | Number of source documents contributing to the candidate pool.                                  |
| max_cands          | Maximum number of candidate chunks considered by the selection algorithm.                       |
| $k / n$            | Number of chunks to select / total candidates.                                                  |
| $c_r$              | Compression ratio (Wang et al.): fraction of tokens selected over total.                        |